Introduction to Computer Science and Programming

Labo 15 – Hands-On Practice Lab

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Today…

Today's lab is dedicated to **hands on practice for understanding of fundamental aspects of the discipline and to verify the students' understanding in coding.**

This lab will help to prepare for next evaluated lab.

- Covered topics in the today evaluated lab:
  - **Theoretical aspects explained during the lessons**
  - **Exercise and practical aspects:**
    - Functions and Arrays
    - Strings
    - Structures
    - Pointers

Hands on Practice Lab – Instructions and Rules

- ***Given the following C++ programs, explain what happens if you try to compile them with a C++ compiler (for example, the one used in class) and execute them providing the inputs that will be given next to the "INPUT:" tag.***
- **Next to the "OUTPUT:" tag, write the expected result after compilation and execution (clearly indicating if the program compiles). So, next to OUTPUT one might write: Compile time error or the result of the execution (which could also be a runtime error, such as dividing a number by zero).**

Exercise 1:

```
#include <iostream>
using namespace std;

void printArray(int A[], int arraySize);
void p(int A[], int arraySize);

int main() {
    int n, numbers[4];
    std::cout << "Enter four numbers: ";
    for(int i = 0; i < 4; i++) {
        std::cin >> numbers[i];
    }
    p(numbers, 4);
    printArray(numbers, 4);
    return 0;
}

void printArray(int A[], int arraySize) {
    for (int i = 0; i < arraySize; i++) {
        cout << A[i] << " ";
    }
}

void p(int B[], int arraySize) {
  B[3] = 9;
}
```

INPUT: 1 2 4 5

OUTPUT:

```cpp
#include <iostream>
using namespace std;

void mistery(int list[], int newList[], int size) {
    for (int i = 0, j = size - 1; i < size; i++, j--) {
        if (list[i]%2==0)
            newList[j] = list[i];
        else
            newList[j] = 0;
    }
}

void printArray(int list[], int size) {
    for (int i = 0; i < size; i++)
        cout << list[i] << " ";
}

int main() {
    const int SIZE = 3;
    int A[SIZE] = {1};
    cin>>A[0]; cin>>A[1];
    int B[SIZE];
    mistery(A, B, SIZE); printArray(B, SIZE);
    return 0;
}
```

Exercise 2:

- **INPUT: 5 8**
- **OUTPUT:**

```cpp
#include <iostream>
using namespace std;

int y = 0;

void t1(int &x);
void t2(int x);
void t3(int &counter);

int main() {
    int x, counter = 0;
    cout << "Insert x = "; cin >> x;
    cout << "Insert y = "; cin >> y;
    t1(x); t2(x); t3(counter); t3(counter); t3(counter);
    cout << x << y << counter << endl;
    return 0;
}

void t1(int &x) {
    x++; y++;
}

void t2(int x) {
    x++;
}

void t3(int &counter) {
    counter++;
}
```

Exercise 3:

- **INPUT: 4 6**
- **OUTPUT:**

```cpp
#include <iostream>
#include <cstring>
#include <string>
using namespace std;

void f1(char str[]);
void f2(string str, char cstr[]);

int main() {
    char cstr[20]; string cppstr;
    cin>>cstr; cin>>cppstr;
    f1(cstr);
    cout << cstr << endl;
    char newCstr[20];
    f2(cppstr, newCstr);
    cout << newCstr << endl;
    return 0;
}

void f1(char str[]) {
    strcpy(str, "Goodbye");
}

void f2(string str, char cstr[]) {
    strcpy(cstr, str.c_str());
}
```

Exercise 4:

- **INPUT: John Doe**
- **OUTPUT:**

Instructions and Rules – Pay attention

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Exercise 5

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    int* arr = new int[n];

    for (int i = 0; i < n; i++) {
        cin >> *(arr + i);
    }

    int* first = arr;
    int* last = arr + n - 2;
    int temp = *first;
    *first = *last;
    *last = temp;

    for (int i = 0; i < n; i++) {
        cout << *(arr + i) << " ";
    }

    return 0;
}
```

- **INPUT: 5 5 4 3 2 1**
- **OUTPUT:**

Exercise 6

```cpp
#include <iostream>
using namespace std;

int main() {
    int x, y;
    int *p;

    cin >> x;
    p = &x;
    *p = 20;

    cin >> y;
    p = &y;
    *p = 50;

    int t = *p;
    *p = 100;

    cout << "x = " << x << ", y = " << y << ", t = " << t << endl;

    return 0;
}
```

- **INPUT: 3 8**
- **OUTPUT:**